`qqbar_nX` implementation notes: local two-loop initial-state collinear subtraction for $d\bar{d} \to HHH$

Generated implementation documentation

Updated 12 May 2026

Abstract

This document records the current implementation of the `qqbar_nX` process in `pygloop`, focusing on the phase-space symmetrized two-loop amplitude

$$d(p_0)\,\bar{d}(p_1) \to H(p_2)\,H(p_3)\,H(p_4).$$

The selected diagrams contain a massive top-quark pentagon loop connected to the incoming massless quark line by two gluons. The local subtraction follows the two initial-state collinear projectors described in Ref. [1], but no UV subtraction is generated here: the heavy subgraph is a pentagon and is UV finite for the present purpose.

The current DOT encoding writes the original graphs and their local IR counterterms as separate graphs sharing a `group_id`. Counterterm numerators are stored globally as spenso tensor networks, while the CT topology is used to expose the denominator structure to GammaLoop's CFF/LTD machinery. This note gives the exact four-dimensional representation of all four selected symmetrized original graphs and all eight associated counterterms, including the loop-momentum routing of every denominator.

Contents

1 Current implementation state

The relevant implementation files are:

`config.toml` `src/processes/qqbar_nX/config.toml`. User-facing process configuration.

`qqbar_nX.py` `src/processes/qqbar_nX/qqbar_nX.py`. Process class, GammaLoop steering, run-card generation, and tests.

`qqbar_nX_graphs.py`
`src/processes/qqbar_nX/qqbar_nX_graphs.py`. DOT parsing, graph selection, and canonical topology grouping.

`qqbar_nX_counterterms.py`
`src/processes/qqbar_nX/qqbar_nX_counterterms.py`. Construction of ISR counterterm DOT graphs.

`qqbar_nX_4d.py`
`src/processes/qqbar_nX/qqbar_nX_4d.py`. Direct four-dimensional Symbolica/spenso evaluator.

The current default artifacts are:

subtracted DOT
`outputs/dot_files/qqbar_nX/qqbar_nX_d_dbar_h_h_h_2L_top_pentagon_isr_subtracted.dot`.

run card `outputs/dot_files/qqbar_nX/qqbar_nX_d_dbar_h_h_h_2L_top_pentagon_isr_subtracted.toml`.

manifest `outputs/dot_files/qqbar_nX/qqbar_nX_d_dbar_h_h_h_2L_top_pentagon_isr_subtracted.manifest.json`.

The current default configuration is:

```
[generation]
symmetrize_final_states = true

[counterterms]
build_isr = true
projector_mode = "leading"
denominator_strategy = "contract"
auxiliary_denominator_mode = "exact_xi_topology"
global_phase = "-i"
uv_inert_dod = -100
use_parametric_xi = false

[standalone_run_card]
enable_threshold_subtraction = true
```

For the symmetrized HHH final state the manifest records

$$N_{\text{original}} = 4, \qquad N_{\text{CT}} = 8, \qquad N_{\text{total}} = 12.$$

The selected original graphs are `GL05`, `GL06`, `GL12`, and `GL13`. Each original graph has two local counterterms:

$$G_{\text{isr\_p1\_ct}} \quad \text{for the } p_0\text{-collinear gluon bridge}, \qquad G_{\text{isr\_p2\_ct}} \quad \text{for the } p_1\text{-collinear gluon bridge.}$$

The current groups are

$$\text{group } 0 = \{\texttt{GL05}, \texttt{GL06} \text{ and CTs}\}, \qquad \text{group } 1 = \{\texttt{GL12}, \texttt{GL13} \text{ and CTs}\}.$$

The original graph remains the group master; the CT graphs are marked `is_group_master=false`. All CT node and edge `dod` attributes are forced to -100, which prevents GammaLoop's own local UV machinery from acting on these artificial local IR counterterms.

2 Generation, selection, and grouping

The raw generation command built by `qqbar_nX.build_feyngen_command` is conceptually

```
generate amp d d~ > h h h | d d~ g h t t~ QED==3 [{2}]
  --only-diagrams
  --global-prefactor-projector '<external spin/color projector>'
  --numerator-grouping no_grouping
  --veto-vertex-interactions V_6 V_9 V_36 V_37
  --number-of-fermion-loops 1 1
  --graph-prefix GL
  --max-multiplicity-for-fast-cut-filter 99
  --symmetrize-final-states true
```

The selector in `qqbar_nX_graphs.py` retains a graph only if it has exactly:

- external particles $d, \bar{d}, H, H, H$;
- two light-quark–gluon vertices `V_74`;
- two top–gluon vertices `V_137`;
- three top–Higgs vertices `V_141`;
- one internal massless light-quark edge joining the two light vertices;
- one gluon bridge from each light vertex to a top–gluon vertex;
- one connected top cycle with five top propagators.

Canonical topology grouping forgets raw node labels and uses label-preserving permutations of the graph. The resulting `group_id` is the minimal unit expected to be locally IR finite after adding the corresponding CTs.

3 Momentum convention and loop-momentum basis

The DOT uses GammaLoop's all-incoming convention. We write

$$p_0 = P(0), \quad p_1 = P(1), \quad p_2 = P(2), \quad p_3 = P(3), \quad p_4 = P(4),$$

and

$$\ell_0 = K(0), \qquad \ell_1 = K(1).$$

For every internal edge e,

$$q_e^\mu = \rho_{e0}\ell_0^\mu + \rho_{e1}\ell_1^\mu + \sum_{r=0}^{4} \eta_{er} p_r^\mu,$$

as stored in the edge attribute `lmb_rep`. The four-dimensional propagator denominator associated with edge e is

$$D_e = q_e^2 - m_e^2 + \mathrm{i}0, \qquad m_e = \begin{cases} m_t, & e \text{ a top edge,} \\ 0, & e \text{ a massless } d \text{ or gluon edge.} \end{cases}$$

The local subtraction depends on the routing of the light-line momentum. The implementation therefore pins the loop momentum basis explicitly:

group	originals	`lmb_id=0`	`lmb_id=1`
0	`GL05`, `GL06`	light bridge edge 10	top edge 5
1	`GL12`, `GL13`	light bridge edge 5	top edge 8

The same basis edges are preserved in every CT graph in the same group. This is important because GammaLoop sets several group-level structures from the group master.

4 Exact four-dimensional integrand representation

For each original or CT graph G, the represented four-dimensional integrand is

$$\mathcal{I}_G(\ell_0, \ell_1) = \frac{\mathcal{N}_G(\ell_0, \ell_1)}{\prod_{e \in E_{\mathrm{int}}(G)} D_e(\ell_0, \ell_1)}, \qquad A_G = \int \frac{\mathrm{d}^4\ell_0}{(2\pi)^4} \frac{\mathrm{d}^4\ell_1}{(2\pi)^4} \mathcal{I}_G(\ell_0, \ell_1). \tag{1}$$

The denominator product is specified graph by graph in Sec. 6.

4.1 Original graph numerators

For original graphs,

$$\mathcal{N}_G = \mathcal{P}_{\mathrm{ext}} \prod_{e \in E_{\mathrm{ext}}} E_{\mathrm{ext}}(e) \prod_{e \in E_{\mathrm{int}}} E_e(q_e) \prod_{v \in V_{\mathrm{int}}} V_v, \tag{2}$$

with the graph-global external projector

$$\mathcal{P}_{\mathrm{ext}} = u(0)\, \bar{v}(1)\, \frac{1}{3}\, \delta_{\mathrm{color}}(\bar{d}, d).$$

The local edge numerator rules exported by GammaLoop are

$$E_d(e : a \to b) = \mathrm{i}\, q_{e,\mu}\, \gamma(s_b, s_a; \mu)\, \delta(c_a, c_b), \tag{3}$$
$$E_t(e : a \to b) = \mathrm{i}\, [q_{e,\mu} \gamma(s_b, s_a; \mu) + m_t\, \delta(s_b, s_a)]\, \delta(c_a, c_b), \tag{4}$$
$$E_g(e : a \to b) = -\mathrm{i}\, g(\mu_a, \mu_b)\, \delta(A_a, A_b), \tag{5}$$
$$E_{\mathrm{ext}}(e) = \mathrm{i} \quad \text{for every external } d, \bar{d}, H \text{ edge.} \tag{6}$$

The vertex numerator rules are

$$V_{74}(qqg) = \mathrm{GC}_{11}\, \gamma(s_L, s_R; \mu_g)\, T(A_g, c_R, c_L), \tag{7}$$
$$V_{137}(ttg) = \mathrm{GC}_{11}\, \gamma(s_L, s_R; \mu_g)\, T(A_g, c_R, c_L), \tag{8}$$
$$V_{141}(ttH) = \mathrm{GC}_{94}\, [P_-(s_L, s_R) + P_+(s_L, s_R)]\, \delta(c_R, c_L). \tag{9}$$

In the independent four-dimensional evaluator $P_- + P_+$ is simplified to the spin identity.

4.2 Counterterm graph numerators

For CT graphs, the final DOT has every local node and edge numerator set to `num` $= 1$. The complete numerator is collected first and then stored in the graph-global `num` attribute as a spenso tensor network. In factorised notation the current default CT numerator is

$$\mathcal{N}_{\text{CT}}(G,b) = \Phi_{\text{CFF}}\,\eta_b\,R_b(G)\,A_b\,B_b(G)^{-1}\,\mathcal{C}_{G,b}[\mathcal{N}_G], \qquad b=1,2. \tag{10}$$

The current default is

$$\Phi_{\text{CFF}} = -\mathrm{i}, \qquad \eta_1 = -\mathrm{i}, \qquad \eta_2 = +\mathrm{i}.$$

The additional phase Φ_{CFF} is a current GammaLoop CFF/event convention phase. It is required for the 3D CFF inspect cancellation. For a pure paper-level four-dimensional Feynman-integrand identity, this phase is set to 1 in a temporary config.

The fixed auxiliary vector is

$$\zeta = p_0 + p_1,$$

and the default implementation uses `use_parametric_xi=false`. The Lorentz prefactors are

$$A_1 = \frac{p_0\cdot\zeta}{p_0\cdot p_1}, \qquad A_2 = \frac{p_1\cdot\zeta}{p_0\cdot p_1}.$$

For massless beams and $\zeta = p_0 + p_1$, both evaluate to one, but the DOT keeps the tensor expression.

Define the Euclidean spatial projection

$$\langle a,b\rangle_3 = a_x b_x + a_y b_y + a_z b_z, \qquad |a|_3^2 = \langle a,a\rangle_3.$$

Let L be the pinned light bridge, g_1 the gluon that is collinear to p_0, and g_2 the gluon that is collinear to p_1. The finite routing fractions are

$$R_1(G) = \frac{\langle q_L, p_1\rangle_3}{\langle q_{g_1}, p_1\rangle_3}, \qquad R_2(G) = \frac{\langle q_L, p_0\rangle_3}{\langle q_{g_2}, p_0\rangle_3}. \tag{11}$$

They deliberately use only spatial components and therefore introduce no loop-energy poles in the graph-global numerator. This was a diagnostic implementation used before the exact auxiliary topology was available; it is not the default subtraction encoded in the current DOT.

The current CFF-compatible auxiliary denominator mode is `exact_xi_topology`. Following the author-confirmed correction of Eqs. (12)–(13) in Ref. [1], the auxiliary propagator is built from the adjacent gluon momentum k_{g_i}, not from the light-bridge momentum k_1. Define

$$A_\xi(q) \equiv (q-\xi)^2 - \xi^2 = q^2 - 2q\cdot\xi. \tag{12}$$

The light-line denominator structures are therefore

$$D_{\text{orig}} = k_1^2\,k_{g1}^2\,k_{g2}^2, \tag{13}$$
$$D_{\Delta_1} = k_1^2\,k_{g1}^2\,A_\xi(k_{g1}), \tag{14}$$
$$D_{\Delta_2} = k_1^2\,A_\xi(k_{g2})\,k_{g2}^2. \tag{15}$$

In DOT form, this is realized by splitting the relevant collinear gluon edge. The original gluon edge id is kept on the unshifted segment so $k_{g_i}^2$ remains an ordinary topology denominator. A new massive segment is inserted on the same path and the active fake in/out helper pair is attached around it; when the helpers are sampled as $Q(2) = Q(4) = \xi$ for Δ_1 or $Q(3) = Q(5) = \xi$ for Δ_2, GammaLoop routes this segment as $k_{g_i} - \xi$. The segment mass is the corresponding helper invariant ξ^2, so the topology denominator is exactly $A_\xi(k_{g_i})$. The opposite gluon denominator is removed by the configured contraction strategy. No ratio or inverse auxiliary denominator is placed in the graph-global numerator in this mode.

The legacy `spatial_proxy` mode removes the opposite gluon propagator from the CT topology and keeps the leading fixed-ζ auxiliary denominator as a loop-energy-free graph numerator factor:

$$B_1(G) = -2(p_0 \cdot p_1)\frac{\langle q_L, p_0\rangle_3}{|p_0|_3^2}, \tag{16}$$

$$B_2(G) = +2(p_0 \cdot p_1)\frac{\langle q_L, p_0\rangle_3}{|p_0|_3^2}. \tag{17}$$

In the current exact topology this expression is no longer part of the default graph numerator; it is retained only as a diagnostic of the old implementation. Denominators with actual loop-energy dependence are part of the graph topology.

The transformation $\mathcal{C}_{G,b}$ acts on the original numerator as:

- replace the internal light-bridge numerator by $\delta_{\rm spin}\delta_{\rm color}$;
- for $b = 1$, replace the p_0-side light vertex by
$$V_{\rm CT,1} = \omega_1\, \mathrm{GC}_{11}\, T(A_{g_1}, c_{\rm ext}, c_{\rm int})\, [2q_{g_1}^{\mu_{g_1}}\,\delta_{\rm spin}];$$
- for $b = 2$, replace the p_1-side light vertex by
$$V_{\rm CT,2} = \omega_2\, \mathrm{GC}_{11}\, T(A_{g_2}, c_{\rm int}, c_{\rm ext})\, [-2q_{g_2}^{\mu_{g_2}}\,\delta_{\rm spin}];$$
- split the same-side collinear gluon edge to introduce the topology denominator $A_\xi(k_{g_i})$;
- contract the opposite gluon edge into the opposite light endpoint, multiplying the opposite light vertex, the opposite gluon numerator, and the corresponding top–gluon vertex into the graph-global numerator.

The orientation signs currently selected from the DOT are:

original graph	ω_1	ω_2
`GL05`, `GL06`	$+1$	-1
`GL12`, `GL13`	-1	$+1$

4.3 Relation to Ref. [1]

The paper writes the light-line factor schematically as

$$D_{\alpha\beta} = \mathrm{i}\,\frac{\bar{v}(p_1)\gamma_\beta(-\not{k}_1)\gamma_\alpha u(p_0)}{k_1^2\, k_{g1}^2\, k_{g2}^2},$$

and constructs two local operators Δ_1, Δ_2 that replace the light-line numerator by the leading collinear numerator and introduce an auxiliary denominator built from a fixed vector. In the published text that auxiliary denominator is written with k_1; the author has confirmed this is a typo and that the intended momenta are k_{g1} and k_{g2}, as encoded in Eq. (15). The `qqbar_nX` code uses the same local leading-collinear replacement logic for the two ISR regions, but differs in two implementation-level ways:

1. the heavy subgraph is a top pentagon with three Higgs insertions;
2. no UV subtraction operator is generated.

5 Original numerator topologies

The original numerator of each selected graph is exactly obtained by applying Eq. (2) and the local rules above to the following directed DOT topologies. An entry $u : h \to v : k$ denotes source node u at hedge h and destination node v at hedge k.

graph	internal nodes	internal edges
GL05	0 V_137, 1 V_137, 2 V_74, 3 V_74, 4 V_141, 5 V_141, 6 V_141	e5 t 0:5 → 1:6; e6 g 0:7 → 3:8; e7 t 6:9 → 0:10; e8 g 1:11 → 2:12; e9 t 1:13 → 5:14; e10 d 3:15 → 2:16; e11 t 5:17 → 4:18; e12 t 4:19 → 6:20.
GL06	0 V_137, 1 V_137, 2 V_74, 3 V_74, 4 V_141, 5 V_141, 6 V_141	e5 t 1:5 → 0:6; e6 g 0:7 → 3:8; e7 t 0:9 → 6:10; e8 g 1:11 → 2:12; e9 t 5:13 → 1:14; e10 d 3:15 → 2:16; e11 t 4:17 → 5:18; e12 t 6:19 → 4:20.
GL12	0 V_74, 1 V_74, 2 V_137, 3 V_137, 4 V_141, 5 V_141, 6 V_141	e5 d 1:5 → 0:6; e6 g 0:7 → 3:8; e7 g 1:9 → 2:10; e8 t 2:11 → 5:12; e9 t 6:13 → 2:14; e10 t 4:15 → 3:16; e11 t 3:17 → 6:18; e12 t 5:19 → 4:20.
GL13	0 V_74, 1 V_74, 2 V_137, 3 V_137, 4 V_141, 5 V_141, 6 V_141	e5 d 1:5 → 0:6; e6 g 0:7 → 3:8; e7 g 1:9 → 2:10; e8 t 5:11 → 2:12; e9 t 2:13 → 6:14; e10 t 3:15 → 4:16; e11 t 6:17 → 3:18; e12 t 4:19 → 5:20.

6 Exact denominator routings

This section gives the exact four-dimensional denominator routing for every graph in the current symmetrized subtracted DOT. For each listed graph,

$$\mathcal{I}_G = \frac{\mathcal{N}_G}{\prod_{e \in S_G} D_e},$$

where S_G is the edge set shown in the corresponding product. For CT graphs, $\mathcal{N}_G$ means the global numerator of Eq. (10) with the original graph and beam indicated by the graph name.

6.1 GL05 and counterterms

GL05.

$$\mathcal{I}_{\texttt{GL05}} = \frac{\mathcal{N}_{\texttt{GL05}}}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11} D_{12}}.$$

edge	part.	q_e	basis
5	t	ℓ_0	lmb_id=1
6	g	$-\ell_0 - p_0 - p_1 + \ell_1 + p_3$	
7	t	$-p_0 - p_1 + \ell_1 + p_3$	
8	g	$\ell_0 - \ell_1 - p_3$	
9	t	$\ell_1 + p_3$	
10	d	$-\ell_0 - p_1 + \ell_1 + p_3$	lmb_id=0
11	t	ℓ_1	
12	t	$-p_0 - p_1 + \ell_1 + p_2 + p_3$	

`GL05_isr_p1_ct`. This is $N_{\mathrm{CT}}(\mathtt{GL05}, 1)$. It keeps $g_1 = e6$ and removes the opposite gluon $e8$ from the active denominator topology:

$$\mathcal{I}_{\mathtt{GL05\_isr\_p1\_ct}} = \frac{\mathcal{N}_{\mathrm{CT}}(\mathtt{GL05}, 1)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	t	ℓ_0	`lmb_id=1`
6	g	$-\ell_0 - p_0 - p_1 + \ell_1 + p_3$	
7	t	$-p_0 - p_1 + \ell_1 + p_3$	
8	t	$-p_0 - p_1 + \ell_1 + p_2 + p_3$	
9	t	$\ell_1 + p_3$	
10	d	$-\ell_0 - p_1 + \ell_1 + p_3$	`lmb_id=0`
11	t	ℓ_1	

`GL05_isr_p2_ct`. This is $N_{\mathrm{CT}}(\mathtt{GL05}, 2)$. It keeps $g_2 = e8$ and removes the opposite gluon $e6$:

$$\mathcal{I}_{\mathtt{GL05\_isr\_p2\_ct}} = \frac{\mathcal{N}_{\mathrm{CT}}(\mathtt{GL05}, 2)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	t	ℓ_0	`lmb_id=1`
6	t	$-p_0 - p_1 + \ell_1 + p_2 + p_3$	
7	t	$-p_0 - p_1 + \ell_1 + p_3$	
8	g	$\ell_0 - \ell_1 - p_3$	
9	t	$\ell_1 + p_3$	
10	d	$-\ell_0 - p_1 + \ell_1 + p_3$	`lmb_id=0`
11	t	ℓ_1	

6.2 `GL06` and counterterms

`GL06`.

$$\mathcal{I}_{\mathtt{GL06}} = \frac{\mathcal{N}_{\mathtt{GL06}}}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11} D_{12}}.$$

edge	part.	q_e	basis
5	t	ℓ_0	`lmb_id=1`
6	g	$\ell_0 - \ell_1 - p_0 - p_1 + p_3$	
7	t	$\ell_1 + p_0 + p_1 - p_3$	
8	g	$-\ell_0 + \ell_1 - p_3$	
9	t	$\ell_1 - p_3$	
10	d	$\ell_0 - \ell_1 - p_1 + p_3$	`lmb_id=0`
11	t	ℓ_1	
12	t	$\ell_1 + p_0 + p_1 - p_2 - p_3$	

`GL06_isr_p1_ct`.

$$\mathcal{I}_{\mathtt{GL06\_isr\_p1\_ct}} = \frac{\mathcal{N}_{\mathrm{CT}}(\mathtt{GL06}, 1)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	t	ℓ_0	`lmb_id=1`
6	g	$\ell_0 - \ell_1 - p_0 - p_1 + p_3$	
7	t	$\ell_1 + p_0 + p_1 - p_3$	
8	t	$\ell_1 + p_0 + p_1 - p_2 - p_3$	
9	t	$\ell_1 - p_3$	
10	d	$\ell_0 - \ell_1 - p_1 + p_3$	`lmb_id=0`
11	t	ℓ_1	

`GL06_isr_p2_ct`.

$$\mathcal{I}_{\texttt{GL06\_isr\_p2\_ct}} = \frac{\mathcal{N}_{\rm CT}(\texttt{GL06}, 2)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	t	ℓ_0	`lmb_id=1`
6	t	$\ell_1 + p_0 + p_1 - p_2 - p_3$	
7	t	$\ell_1 + p_0 + p_1 - p_3$	
8	g	$-\ell_0 + \ell_1 - p_3$	
9	t	$\ell_1 - p_3$	
10	d	$\ell_0 - \ell_1 - p_1 + p_3$	`lmb_id=0`
11	t	ℓ_1	

6.3 `GL12` and counterterms

`GL12`.

$$\mathcal{I}_{\texttt{GL12}} = \frac{\mathcal{N}_{\texttt{GL12}}}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11} D_{12}}.$$

edge	part.	q_e	basis
5	d	$-\ell_0 + p_0$	`lmb_id=0`
6	g	$-\ell_0 + p_0 + p_1$	
7	g	ℓ_0	
8	t	$\ell_1 + p_0 + p_1 - p_2$	`lmb_id=1`
9	t	$-\ell_0 + \ell_1 + p_0 + p_1 - p_2$	
10	t	ℓ_1	
11	t	$-\ell_0 + \ell_1 + p_0 + p_1$	
12	t	$\ell_1 + p_0 + p_1 - p_2 - p_3$	

`GL12_isr_p1_ct`. This keeps $g_1 = e7$ and removes $e6$:

$$\mathcal{I}_{\texttt{GL12\_isr\_p1\_ct}} = \frac{\mathcal{N}_{\rm CT}(\texttt{GL12}, 1)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	d	$-\ell_0 + p_0$	`lmb_id=0`
6	t	$\ell_1 + p_0 + p_1 - p_2 - p_3$	
7	g	ℓ_0	
8	t	$\ell_1 + p_0 + p_1 - p_2$	`lmb_id=1`
9	t	$-\ell_0 + \ell_1 + p_0 + p_1 - p_2$	
10	t	ℓ_1	
11	t	$-\ell_0 + \ell_1 + p_0 + p_1$	

`GL12_isr_p2_ct`. This keeps $g_2 = e6$ and removes $e7$:

$$\mathcal{I}_{\texttt{GL12\_isr\_p2\_ct}} = \frac{\mathcal{N}_{\mathrm{CT}}(\texttt{GL12}, 2)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	d	$-\ell_0 + p_0$	`lmb_id=0`
6	g	$-\ell_0 + p_0 + p_1$	
7	t	$\ell_1 + p_0 + p_1 - p_2 - p_3$	
8	t	$\ell_1 + p_0 + p_1 - p_2$	`lmb_id=1`
9	t	$-\ell_0 + \ell_1 + p_0 + p_1 - p_2$	
10	t	ℓ_1	
11	t	$-\ell_0 + \ell_1 + p_0 + p_1$	

6.4 `GL13` and counterterms

`GL13`.

$$\mathcal{I}_{\texttt{GL13}} = \frac{\mathcal{N}_{\texttt{GL13}}}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11} D_{12}}.$$

edge	part.	q_e	basis
5	d	$-\ell_0 + p_0$	`lmb_id=0`
6	g	$-\ell_0 + p_0 + p_1$	
7	g	ℓ_0	
8	t	$\ell_1 - p_0 - p_1 + p_2$	`lmb_id=1`
9	t	$\ell_0 + \ell_1 - p_0 - p_1 + p_2$	
10	t	ℓ_1	
11	t	$\ell_0 + \ell_1 - p_0 - p_1$	
12	t	$\ell_1 - p_0 - p_1 + p_2 + p_3$	

`GL13_isr_p1_ct`. This keeps $g_1 = e7$ and removes $e6$:

$$\mathcal{I}_{\texttt{GL13\_isr\_p1\_ct}} = \frac{\mathcal{N}_{\mathrm{CT}}(\texttt{GL13}, 1)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	d	$-\ell_0 + p_0$	`lmb_id=0`
6	t	$\ell_1 - p_0 - p_1 + p_2 + p_3$	
7	g	ℓ_0	
8	t	$\ell_1 - p_0 - p_1 + p_2$	`lmb_id=1`
9	t	$\ell_0 + \ell_1 - p_0 - p_1 + p_2$	
10	t	ℓ_1	
11	t	$\ell_0 + \ell_1 - p_0 - p_1$	

`GL13_isr_p2_ct`. This keeps $g_2 = e6$ and removes $e7$:

$$\mathcal{I}_{\texttt{GL13\_isr\_p2\_ct}} = \frac{\mathcal{N}_{\mathrm{CT}}(\texttt{GL13}, 2)}{D_5 D_6 D_7 D_8 D_9 D_{10} D_{11}}.$$

edge	part.	q_e	basis
5	d	$-\ell_0 + p_0$	`lmb_id=0`
6	g	$-\ell_0 + p_0 + p_1$	
7	t	$\ell_1 - p_0 - p_1 + p_2 + p_3$	
8	t	$\ell_1 - p_0 - p_1 + p_2$	`lmb_id=1`
9	t	$\ell_0 + \ell_1 - p_0 - p_1 + p_2$	
10	t	ℓ_1	
11	t	$\ell_0 + \ell_1 - p_0 - p_1$	

7 Four-dimensional evaluator

The independent 4D test path in `qqbar_nX_4d.py` constructs

$$\mathcal{I}_{\mathcal{G}}^{4D} = \frac{\mathrm{Contr}\left[\mathcal{P}_{\mathrm{ext}}\,\mathcal{N}_{\mathcal{G}}^{\mathrm{graph}} \prod_v \mathcal{N}_v \prod_e \mathcal{N}_e\right]}{\prod_{e\in E_{\mathrm{int}}\setminus E_{\mathrm{dummy}}} \left(Q_e(0)^2 - Q_e(1)^2 - Q_e(2)^2 - Q_e(3)^2 - m_e^2\right)}.$$

It replaces `spenso::projm`+`spenso::projp` by the Dirac metric, simplifies color with Symbolica community, and contracts the tensor network with spenso before numerical evaluation. If a graph-global numerator contains $\mathrm{OSE}(e)$, it is replaced in the 4D diagnostic by

$$\mathrm{OSE}(e) \to \sqrt{Q_e(1)^2 + Q_e(2)^2 + Q_e(3)^2 + m_e^2}.$$

Implementation caveat: the current DOTs do not write an explicit top mass attribute on top edges. The formal GammaLoop interpretation uses $m_t = \mathrm{MT}$ for top denominators; the current independent 4D evaluator only subtracts masses present as DOT edge attributes. The tested collinear identity is nevertheless meaningful because the heavy-loop denominators are finite in the collinear approach and the same DOT object is evaluated for the originals and their CTs. A later cleanup should infer model masses by particle name.

8 Collinear tests and current results

The process-specific tests are exposed through:

```
PYTHONPATH=src .venv/bin/python src/pygloop.py --process qqbar_nX test --mode 4D
PYTHONPATH=src .venv/bin/python src/pygloop.py --process qqbar_nX test --mode
    gammaloop
```

The current test configuration uses $x = 0.3$, precision `ArbPrec`, and

$$|k_\perp| = \lambda\sqrt{s}, \qquad \lambda \in \{1, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\}.$$

For every λ and group, the diagnostic ratio is

$$\rho_{\mathrm{group}} = \frac{\left|\sum_{i\in\mathrm{group}} \mathcal{I}_i\right|}{\sum_{i\in\mathrm{group}} |\mathcal{I}_i|}.$$

A local cancellation of the leading collinear singularity appears as a ratio that decreases linearly with λ.

8.1 Pure 4D paper-level diagnostic, $\Phi_{\mathrm{CFF}} = 1$

The independent 4D algebra check should be run with the CT global phase reset to 1, because it tests the paper-level Feynman-integrand identity, not the GammaLoop 3D CFF event convention:

```
tmp=/tmp/qqbar_nX_4d_phase1.toml
cp src/processes/qqbar_nX/config.toml "$tmp"
perl -0pi -e 's/global_phase = "-i"/global_phase = "1"/' "$tmp"
PYTHONPATH=src .venv/bin/python src/pygloop.py --process qqbar_nX \
  --qqbar-nx-config "$tmp" \
  --overwrite-process-basename qqbar_nX_4dphase1 \
  --m_top 1500 test --mode 4D
```

The current results are:

beam	group	$\lambda = 10^{-3}$	$\lambda = 10^{-5}$	$\lambda = 10^{-7}$
p_0	0	$1.207432422884 \times 10^{-2}$	$1.219429826526 \times 10^{-4}$	$1.219547309408 \times 10^{-6}$
p_0	1	$5.953650138218 \times 10^{-3}$	$5.918551126568 \times 10^{-5}$	$5.918194450500 \times 10^{-7}$
p_1	0	$5.110458964993 \times 10^{-3}$	$5.133079576218 \times 10^{-5}$	$5.133298939401 \times 10^{-7}$
p_1	1	$5.329260932463 \times 10^{-3}$	$5.304017780889 \times 10^{-5}$	$5.303763375967 \times 10^{-7}$

This confirms that the 4D numerator/projector construction cancels locally.

8.2 Default 4D diagnostic with $\Phi_{\mathrm{CFF}} = -\mathrm{i}$

With the default CFF phase, the same 4D Feynman-integrand diagnostic does not cancel, as expected:

beam	group	$\rho_{\mathrm{group}}(\lambda = 10^{-7})$
p_0	0	$6.932501873977 \times 10^{-1}$
p_0	1	$7.070605995487 \times 10^{-1}$
p_1	0	$5.417365569681 \times 10^{-1}$
p_1	1	$7.068718138865 \times 10^{-1}$

This is a convention check: the $-\mathrm{i}$ phase is inserted for the GammaLoop CFF representation, not for the pure 4D algebra identity.

8.3 GammaLoop 3D CFF inspect, $m_t = 1500$

The non-threshold-clean heavy-mass test was run with:

```
PYTHONPATH=src .venv/bin/python src/pygloop.py --process qqbar_nX \
  --m_top 1500 --clean test --mode gammaloop
```

The current results are:

beam	group	$\lambda = 10^{-3}$	$\lambda = 10^{-5}$	$\lambda = 10^{-7}$
p_0	0	$2.156437298117 \times 10^{-3}$	$2.162682427278 \times 10^{-5}$	$2.162745021408 \times 10^{-7}$
p_0	1	$1.019780744626 \times 10^{-2}$	$1.015716257667 \times 10^{-4}$	$1.015675267491 \times 10^{-6}$
p_1	0	$2.156437298117 \times 10^{-3}$	$2.162682427278 \times 10^{-5}$	$2.162745021408 \times 10^{-7}$
p_1	1	$1.019780744626 \times 10^{-2}$	$1.015716257667 \times 10^{-4}$	$1.015675267491 \times 10^{-6}$

The p_0 and p_1 summaries are identical in the current fractional test because the GammaLoop inspect path solves the same pinned light-edge ray. The important result is the linear decrease with λ in each group.

8.4 GammaLoop 3D CFF inspect, physical $m_t = 173$

The physical-mass test was run at $\sqrt{s} = 1000$ GeV with threshold subtraction enabled:

```
PYTHONPATH=src .venv/bin/python src/pygloop.py --process qqbar_nX \
  --overwrite-process-basename qqbar_nX_mt173 \
  --m_top 173 --clean test --mode gammaloop
```

The generated threshold overlap structures were active:

$$\text{group 0 sizes } [13, 12, 12], \qquad \text{group 1 sizes } [9, 9, 8, 8].$$

The current results are:

beam	group	$\lambda = 10^{-3}$	$\lambda = 10^{-5}$	$\lambda = 10^{-7}$
p_0	0	$1.388028023261 \times 10^{-3}$	$1.409386585386 \times 10^{-5}$	$1.409618038701 \times 10^{-7}$
p_0	1	$1.738779562355 \times 10^{-3}$	$1.729480510655 \times 10^{-5}$	$1.729359476107 \times 10^{-7}$
p_1	0	$1.388028023261 \times 10^{-3}$	$1.409386585386 \times 10^{-5}$	$1.409618038701 \times 10^{-7}$
p_1	1	$1.738779562355 \times 10^{-3}$	$1.729480510655 \times 10^{-5}$	$1.729359476107 \times 10^{-7}$

This demonstrates the same local cancellation pattern in the physical-mass setup with threshold sectors present.

9 Standalone GammaLoop run card

The generated run card has named command blocks and no non-empty top-level `commands` block, so loading the TOML does not automatically run generation or inspection. The intended standalone sequence is:

```
GL=/Users/vjhirsch/Documents/Work/gammaloop_dev/target/dev-optim/gammaloop
CARD=outputs/dot_files/qqbar_nX/qqbar_nX_d_dbar_h_h_h_2L_top_pentagon_isr_subtracted.
    toml
STATE=outputs/gammaloop_states/
    qqbar_nX_standalone_qqbar_nX_d_dbar_h_h_h_2L_top_pentagon_isr_subtracted

$GL --clean-state "$CARD"
$GL -o -s "$STATE" run generate_subtracted_integrand
$GL -o -s "$STATE" run inspect_collinear_p1
$GL -o -s "$STATE" run inspect_collinear_p1_arb
$GL -o -s "$STATE" run inspect_collinear_p2
$GL -o -s "$STATE" run inspect_collinear_p2_arb
$GL -o -s "$STATE" run low_stat_integrate_pm
$GL -o -s "$STATE" run low_stat_integrate_pp
```

The helper shell script supports sectioned execution, including an `open` mode that drops into the live GammaLoop CLI.

10 Current caveats and next cleanup items

1. The independent 4D evaluator should infer masses from the model or particle name, not only from explicit DOT edge `mass` attributes.
2. The parametric-ξ option is implemented as configuration and additional-parameter plumbing, but the current validated results use the fixed $\zeta = p_0 + p_1$ setup.
3. The current GammaLoop inspect p_0 and p_1 summaries use the same pinned light-edge fractional ray. This is a useful grouped regression test, but a later diagnostic should also print independent solved rays for the two beam limits.

References

[1] D. Kermanschah and M. Vicini, *Heavy-quark box-loop corrections to $q\bar{q} \to Z\gamma$ at two loops in QCD*, arXiv:2603.03169 [hep-ph], https://arxiv.org/abs/2603.03169.